SECURITY ASPECTS OF THE CURRENT UNITED NATIONS SPACE LAW AGENDA +

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The Current Agenda

The current space law agenda of the United Nations is largely a familiar one. Four items on the formal agenda are before the UN Committee on the Peaceful Uses of Outer Space (COPUOS) and its Legal Subcommittee. The subject of arms control in outer space, which was referred in 1981 to the 40 member Committee on Disarmament,[1] is also before these two committees.

A few countries opposed to military uses of outer space, even for purposes other than deployment of weaponry, have been raising with some insistence an issue broader than the dangers of an arms race in outer space. This is the issue of outer space militarization, which has not yet been placed on the UN's formal space law agenda. Even though the relationship of arms control and militarization of space to security in the classical sense is evident and of great importance, security implications are not always of predominance on the outer space law agenda. The current agenda consists of the following four items:

> (i) legal implications of remote sensing of the earth from outer space, with the aim of formulating draft principles;
> (ii) consideration of the possiblity of supplementing the norms of international law relevant to the use of nuclear power sources (NPS) in outer space;
> (iii) matters relating to the definition and/or delimitation of outer space and outer space activities, bearing in mind, *inter alia,* questions relating to the geostationary orbit; and
> (iv) elaboration of draft principles governing the use by states of artificial earth satellites for direct television broadcasting (DBS).

This article sketches the disposition of three of these issues, which are remote sensing, NPS, and the definition and/or delimitation of outer space. Since DBS is a security issue only in the view of closed societies which fear the free flow of information and ideas, the legal regulation of the geostationary orbit, and outer space arms control, and since others are addressing this topic, this article does not deal with it.

Remote Sensing

Remote sensing remains a priority item of the Legal Subcommittee agenda. In fact, it is currently the only priority item. Despite the fact that this is a priority item and

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[1]This body is the principal international forum for the consideration of multilateral arms control and disarmament agreements.

eleven meetings of the 1982 Legal Subcommittee's working group have been held, no new areas of consensus have been achieved and no brackets have been removed from the heavily bracketed subcommittee working text.[2]

The principal issues in this negotiation are: the scope of the principles; whether the remotely sensed state must consent before data or certain categories of data obtained from space may be disseminated; what access to data must remote sensing states provide others; and what responsibility, if any, will states have for the dissemination and use on earth of data remotely sensed from space.

The principal concerns which have been expressed by the majority of participants in the remote sensing negotiations have been related to the exploitation of natural resources. Some COPUOS members have expressed this anxiety in ideological terms, asserting that "permanent sovereignty over natural resources" includes sovereignty over data on those resources gathered from outer space and information derived from such remote sensing data.

The Soviet Union has supported this and has asserted as well that states have an obligation not to use or disseminate remote sensing data for use against the interests of any other state. The latter is a particularly pernicious suggestion, despite the surface appeal it has for some, since, if adopted, it would end open dissemination of data by making the sensing state responsible for the use to which others put data received. Further, endless disputes may arise over what is contrary to the interests of another state, a standard which is as unrealistic as it is unworkable. The Soviet Union would also restrict non-consensual dissemination of fine resolution data, while permitting dissemination of gross resolution data such as that derived from meteorological satellites. This restriction would be accomplished by the latest Soviet proposal that continues to draw the line at a spatial resolution of 50 meters,[3] even though they have indicated that the actual figure is open to negotiation.

The French also support some such restrictions. Their representatives profess to see no conflict between doing so and proceeding with their commercial Spot remote sensing program with its finer 10 meter resolution.

The United States (operating a Landsat program with 30 meter resolution) has responded that sovereignty over resources does not include sovereignty over information about those resources obtained from a platform in space any more than it does information obtained by any other means. The United States further asserts that the principles under consideration would not affect the freedom to gather information from space; such restrictions on dissemination would serve only to enforce a monopoly of information by the states with remote sensing systems, end the open Landsat type system with its public data centers and direct readout to foreign ground stations, and impair drastically the continuity and utility of remote sensing data.

The United States has pointed out that information obtained from remote sensing is used for the study of problems and phenomena which do not necessarily correspond to the contours of an individual state. Furthermore, the United States has expressed great

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[2]The text can be found in the Legal Subcommittee's 1981 and 1982 reports, UN Doc. A/AC.105/288, Annex I, appendix, and A/AC.105/305, Annex I, appendix.

[3]USSR Working Paper, WG/RS (1982) WP.4, 5 February 1982, A/AC.105/305, Annex I, p. 18.

skepticism about concerns that dissemination of remotely sensed data without sensed state consent would impair the ability of the sensed state to protect its rights over the timing, terms and conditions of the exploitation of its natural resources.

This point has been developed by the United States in a different context. During the 1981 Legal Subcommittee session the Soviet Union's delegate theorized that exporters' wheat prices could be effected by information obtained and disseminated about a poor Soviet wheat crop. The U.S. responded that this suggestion fails, since wheat prices would increase anyway when a country with a poor wheat crop conceals such information and then attempts to make up its shortfall on the international market. The cost of failure is merely shifted to the more successful growers and their traditional foreign buyers and domestic consumers. Furthermore, the information genie cannot be put back in the bottle with such a tremendous amount of information now available from non-space and space sources. Not only would very few countries be in a position to benefit from restricting dissemination of space derived crop information, but the attempt to restrict rather than to improve the international supply and demand information base is not worthy of protection by developing space law principles.

The foregoing may have contributed to a significant shift which has begun to occur on this issue in favor of the position that sovereignty over natural resources does not include sovereignty over information obtained about or from those resources. More important factors contributing to this significant shift are growth in state participation in the Landsat program through regional ground stations, and investment in use of remote sensing data. One Latin representative questioned whether the Third World had properly assessed its real interests in asserting that the right to restrict dissemination of data from space was part of permanent sovereignty.[4] An African representative stated that this point should be taken into account as the number of sensing states increases. In addition, further erosion of the previously united front on the issue occurred during the 1982 Legal Subcommittee session when Brazil introduced a package proposing the deletion of the prior consent principle (Principle XV)[5] entirely, together with the strengthening of the principle on access by sensed states to remotely sensed data (Principle XII).[6] While the Soviet Union continued to assert that prior consent must be obtained before surveillance of a state's natural resources, there was a noticeable shift of emphasis on the part of many delegations from restricting dissemination to gaining guaranteed access to remotely sensed data.

While open and non-discriminatory access to civil remote sensing data has been and remains United States policy, the United States is in the process of working out a transfer of operational civil land remote sensing to the private sector. It is not altogether clear what implications commercial operation will or should have for that policy.

The issue of high resolution data does have another more direct security aspect. One facet of the issue was emphasized in political terms at the 1982 Legal Subcommittee

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[4]This was done despite the fact that a Mexican comprehensive draft set of principles introduced in that session had included an unqualified prior consent requirement. Mexico: Working Paper, WG/RS (1981)/WP.2, 19 March 1981, Report of the Legal Subcommittee, Doc. A/AC.105/288, Annex I, p. 13.

[5]Brazil: Working Paper, WG/RS (1982)/WP.11, 8 February 1982, Report of the Legal Subcommittee, A/AC.105/305, Annex I, p. 20.

[6]*Id.*

session when the Soviet Union's representatives repeatedly adverted to the damage which could be caused to the security of countries in the Middle East if high resolution remote sensing data were provided Israel. There is, of course, an undeniable military importance to surveillance of the earth's surface, and this would be no less true for high resolution surveillance from space than from drones or high altitude aircraft.

The Soviet Union asserts, however, implausibly, that the remote sensing principles are to be broad in scope and cover all remote sensing, which by implication includes any national security programs. Their definition, which is purely technical, would not distinguish between programs of different purposes or nature. They have not yet accepted the United States' view that the draft principles can only apply to civil remote sensing such as our Landsat or the French Spot systems.

To attempt to draft remote sensing principles to cover anything beyond civil remote sensing would immediately make it impossible to draft acceptable language for a good many of the matters now included in the draft. For example, the draft now has principles on such matters as making available to sensed states opportunities to participate in remote sensing programs and the duty to provide sensed states with timely and non-discriminatory access to the remotely sensed data. Can anyone seriously envisage international consensus on a principle that the states conducting surveillance must allow the sensed state an opportunity to participate in the program and have access to the result? Moreover, such a principle, if adopted, would probably have detrimental effects in the area of arms control, where it is generally considered an important safeguard against violation that each party can never be quite sure precisely how good the other party's national technical means of verification are. Further, a ban on sharing high resolution remote sensing data and information with other states, including allies, without the sensed state's consent would be not only unrealistic, but completely unverifiable. In fact, the application of the proposed principles to anything other than civil remote sensing is so implausible that one would hesitate to take the ostensible Soviet position as either literal or immutable. Rather, it is possible that the purely technical and inclusive Soviet proposed definition has led them to include a certain number of potential loopholes in the various draft principles allowing them a means of frustrating any unwanted efforts to apply the principles to gain access to closed Soviet systems. Such an approach does not seem as promising as to define the scope of the principles straightforwardly as applying to civil remote sensing.

In assessing the security implications of the dissemination of high resolution remote sensing data, one has to bear in mind the increasing numbers of countries which will possess both the technology and the means to carry out sophisticated remote sensing programs. A ban on dissemination would only deny information to countries or groups lacking either their own capability or access to data from a sensing state willing secretly to violate the principle. Further, what principle would justify barring a state without the means to maintain its own remote sensing system from obtaining data about a potential or actual adversary who possesses such a system?

There is a variant on this issue which raises somewhat different considerations, that is, the impact on security of states from the routine public dissemination of high resolution data from some future civil system with very high resolution. Would this routine availability of data provide significantly better targeting data for and heighten the danger of aggression, armed attack or sabotage in various parts of the world by non-remote sensing states, political terrorist groups, or lunatics? Are there civil

requirements which make the operation of such high resolution systems likely? If so, are the risks associated with them sufficient to offset the benefits and justify constraints of some sort? Further, even if the potential risks were deemed significant, would the appropriate constraint be an international norm prohibiting public disclosure without sensed state consent? I do not propose to answer these questions, but offer them for reflection.

Nuclear Power Sources

This topic was thrust onto the international space law agenda when a Soviet satellite carrying a nuclear power reactor, Cosmos 954, malfunctioned and descended over Canada on January 24, 1978. Following that accident, the Canadian government moved to bring the issue of use of nuclear power sources before the United Nations and, in 1978, the General Assembly recommended that the Scientific and Technical Subcommittee establish a working group of experts to deal with "consideration of technical aspects and safety measures relating to the use of nuclear power sources in outer space."[7] In 1979, the Assembly recommended that the Scientific and Technical Subcommittee continue its work and that the Legal Subcommittee include on its agenda an item entitled: "Review of existing international law relevant to outer space activities with a view to determining the appropriateness of supplementing such law with provisions relating to the use of nuclear power sources in outer space."[8]

The Scientific and Technical Subcommittee's expert working group produced a consensus report in 1981,[9] which, together with replies of governments to a Secretariat questionnaire,[10] a series of working papers submitted by Canada in 1980 and thereafter, and papers submitted by other governments in 1981 and 1982, forms the basis of the Legal Subcommittee's current work. The name of the item has evolved and the Subcommittee is now charged with "consideration of the possibility of supplementing" existing international law. That obscure change in mandate, plus the formation of a working group to deal with the matter mark, in UN terms, substantial progress since the Legal Subcommittee began on its work on NPS.

The principal issues under this item now are related to whether international safety standards can and should be articulated; whether those standards are adequately identified in the S & T experts working group report; whether nuclear power reactors, as contrasted with radioisotopic power generators, must be subject to further restriction or actual prohibition in certain cases; whether nuclear power sources require a special liability regime; and whether special rules on notification, including pre-flight notice, and on assistance to states should be articulated. The United States, the Soviet Union and several other countries have noted with approval in the Legal Subcommittee the Scientific and Technical Subcommittee experts working group conclusion that nuclear

[7]UNGA Resolution 33/16.

[8]UNGA Resolution 34/66.

[9]Report of the Scientific and Technical Subcommittee, A/AC. 105/287, Annex II.

[10]UN Doc. A/AC.105/C.2/14 and Add. 1 to 3.

power sources can be used safely in outer space, provided that all necessary safety measures it described are met. Despite this consensus at the technical level, achieved to a large degree through the constructive efforts of the Canadian participants in the experts group, Canada took a different point of departure at the 1981 Legal Subcommittee session; Canada suggested that most missions using NPS did not benefit all the states subjected to risk and that the governing principle for use of NPS should be "no benefit no risk."[11] This was rejected by a number of delegations, not merely the leading space powers, as unrealistic.

The real bone of contention as far as the safety issues are concerned is the use of nuclear power reactors in low earth orbit. That was the nuclear power source involved in the flight of Cosmos 954, and that is the use requiring the most stringent safety precautions. Published sources to which people discussing such esoterica usually refer, such as *Aviation and Space Weekly* and *Defense Daily,* would have it that Cosmos 954 and others in this series are ocean radar surveillance satellites, designed to function for a period of up to two months in low earth orbit before being reboosted to a 600 mile orbit where the reactor's radioactive constituents would have sufficient time to decay to safe levels before re-entry. These same sources had reported the launching of as many as fifteen of these satellites from the start of the program through 1981. Following the failure of the reboost when Cosmos 954 malfunctioned, there was a breakup of the satellite and extensive dispersal of the radioactive material upon re-entry.

With regard to nuclear power reactors, the experts working group had reached consensus on the following:

> If reactors are intended for use in low orbits where radioactive materials do not have sufficient time to decay to an acceptable level, safety depends on the start of operation in orbit and the success of boosting NPS to a higher orbit after operation is completed. In the event of an unsuccessful boost into higher orbit, the system should in all credible circumstances be capable of dispersing the radioactive material so that when the material reaches the earth the radiological situation conforms to the recommendations of ICRP when relevant.

Canada subsequently rejected the use of low earth orbit nuclear power reactors, as it had rejected other aspects of the experts' consensus. In a statement made to the Legal Subcommittee on February 10, 1982, the Canadian representative stated that "the risks created by nuclear reactors in low-earth orbit cannot be sufficiently reduced so as to render them acceptable." Canada challenged the adequacy of dispersal by citing the plating of radioactive material onto components of the re-entering spacecraft which they found in the Cosmos 954 incident. The United States pointed out that designs, such as that of Cosmos 954, which result in significant plating of highly radioactive materials on sizable components surviving re-entry, obviously do not meet the criterion set out in the experts group report, but that this did not detract from the appropriateness of the criterion.

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[11] A Canadian perspective on the NPS issue through the 1981 Legal Subcommittee, including the rationale for the shift away from the experts group consensus, can be found in an article by a member of their delegation, Jason Reiskind, *Toward a Responsible Use of Nuclear Power in Outer Space - The Canadian Initiative in the United Nations,* 6 ANNALS OF AIR AND SPACE L. 461, 1981.

The current Canadian proposal for a safety regime for nuclear power sources in space reflects the shift in Canadian view, albeit with some peculiar wrinkles. The principle features as they relate to nuclear power reactors in low earth orbit are:

1. A general prohibition on the use of nuclear power reactors in low earth orbit;
2. Exceptions may be made only for national security reasons, and then only subject to the following conditions:
   a. it must be announced that an NPS is being used in low earth orbit for national security purposes;
   b. treble damages must be paid for any damage caused;
   c. the number launched per year and in orbit at any time may not exceed a limit (to be set);
   d. the amount of radioactive fuel may not exceed a specified limit; and
   e. there must be two reboost backup systems where it is not planned to bring the object back to earth in a controlled re-entry, and a level of control at least meeting the standard for manned spacecraft is required where the space object is to return to earth at the completion of its mission.

The Canadian proposal does not appear entirely well conceived. Although the United States has only flown one nuclear power reactor, an experimental one in 1965 in earth orbit sufficiently high that the reactor will have essentially no radioactivity when its orbit decays in approximately 4,000 years, nuclear power reactors may be the most promising source for meeting high energy requirements in future space missions. If, for example, mankind is to send scientific missions to Jupiter and beyond, it may have to be done with NPS, both radioisotope power sources and nuclear reactors—and we cannot exclude the possibility that such missions would require the start-up of the reactor in a low earth orbit. Similarly, there may be high energy requirements for other civil applications in low earth orbit, such as industrial processing. If, as the experts had agreed, nuclear reactors can be used safely in space, such proposed missions should be evaluated on a case-by-case basis, taking into account the technical need, the benefits to be derived, and the probable risks associated with the proposal. Like any national security missions, civil missions should be flown only with stringent safety precautions after rigorous safety analysis. But the Canadian proposal would totally rule out these possible civil applications now, *a priori,* while making purported exception for national security missions. Further, the exception the Canadian paper would make for low earth orbit nuclear reactors does not appear to be genuine, since it is hedged with conditions unlikely to be acceptable to agencies responsible for national security activities, and certainly not to those of the Soviet Union. Should such a high energy source be required for future United States missions of national security significance, such missions would, in all probability, require classification and confidentiality precluding the public announcements called for by the Canadian proposal. The suggestion of treble damages is perhaps even more improbable as a candidate for consensus, if not for the financial burden, then for the implication of tortiousness in such use of nuclear power reactors.

The United States continues to support the effort to elaborate an international regime regarding the use of nuclear power sources. It is likely, however, that the consensus achieved in the experts' working group of the Scientific and Technical Subcommittee will be the basis of any consensus on the legal side concerning safety.

Definition and Delimitation of Outer Space

The UN's consideration of the question of definition of outer space is the longest standing item on the agenda of the Legal Subcommittee. It was on the agenda, without mention of delimitation, outer space activities, or the geosynchronous orbit (GSO) at the Seventh Session of the Legal Subcommittee, in Geneva, Switzerland, in 1968. While the literature has been filled with discussions of the question both before and since that time, and it has been discussed and debated year after year in the Legal Subcommittee, it appeared no nearer an answer and no more in need of an answer when the 1981 Legal Subcommittee met.

The two basically opposed approaches to the definition and delimination of outer space are illustrated by Vladimir Kopal[12] and Stanley Rosenfield.[13]

Kopal, now a member of the UN Outer Space Division secretariat, recites the standard arguments for a definition and delimitation of outer space, finding such need in a variety of factors, including the growth of space legislation and the number of space objects, the development of aerospace transport, the prospect of solar power satellites and the equatorial states' claim to the geostationary orbit. He asserts that the lower limit of outer space is the most important line to fix and that a right passage for space objects, including a freedom of transit airspace of other states, should be recognized. In so doing, he suggests that "where sovereignty ends" should be left aside.[14] In addition, he endorses the lowest satellite perigees as the approximate lower limit and unsurprisingly finds the proposal of the Soviet Union to be a well founded suggestion for agreement that a boundary between air space and outer space be no higher than 100/110 km above sea level. In something of a contradiction of his assertion that only the lower limit of space, not the upper limit of sovereignty, be dealt with, Kopal states that placing that lower limit too high would impair space activities since penetration below that limit would require the consent of any subjacent state.

Rosenfield considers that after more than twenty years of space activity, "there is still no need for a specific line distinguishing air space from outer space."[15] He notes that treaties have either avoided the problem or taken the functional approach. He does not consider that the shuttle raises any current problem of delimitation since, on ascent it is like any other launcher and on descent it is not highly maneuverable and its descents have not to date raised the issue.

The urge to define outer space or delimit a boundary between it and airspace does not appear to be driven by any real need. It seems beyond legal challenge that satellites orbiting the earth are operating in outer space and under the outer space legal regime. Proposals to agree that the lower limits of outer space are no higher than that would clarify or add nothing regarding activities in orbit or beyond. Further, the geosynchronous orbit issue which has been attached to this item is extraneous to the

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[12]Kopal, *The Question of Defining Outer Space*, 8 J. SPACE L. 154, 1980.

[13]Rosenfield, *Where Air Space Ends and Outer Space Begins*, 7 J. SPACE L. 136, 1979.

[14]Kopal, *supra* at note 12.

[15]Rosenfield, *supra* at note 13.

boundary question since, whatever the scientific, logical or legal basis for the peculiar claim of equatorial state sovereignty over that orbit, not even the equatorial states claim that the GSO is part of airspace. To date, no space system operation, including the return of the United States shuttle orbiter, has raised a problem requiring definition or delimitation. Nor has anyone identified a credible security advantage to states from setting a boundary at the limits usually discussed. I should point out, in passing, that the proposal of the Soviet Union, usually cited as favoring establishment of a boundary at 100 to 110 kms, does not do so. Instead, it proposes in effect that the lower limit of outer space is no higher than that altitude and leaves the issue of the boundary between airspace and outer space and, with it, the upper limit of airspace for future agreement. Thus, the Soviet proposed approach would do nothing to advance the cause of those concerned to know how high they can assert sovereignty and, in their view, protect their security interests.

One should be fairly skeptical that a boundary would have significant benefit for subjacent state security or that it could be adopted without disadvantage to the legitimate interests of states in the use of space. Is it reasonable to fear that better surveillance or reconnaissance could occur from the area below the lowest satellite perigees and above the upper limit of flight with air-breathing engines than can be carried out from orbit? If not, what other non-war activity of adverse security impact could be conducted more advantageously there? Or is it feared that the zone might provide advantages in the delivery of destructive force to military targets? For the latter fear, in what way would delimitation help? A legal boundary between airspace and outer space would be devoid of significance between combatants during hostilities. At the same time, to raise a boundary significantly higher than the upper limit of flight powered by air-breathing engines without also agreeing to an unqualified right of passage or transit to and from outer space could inhibit access to space. It could do so with unequal effect, burdening many states more than it would the United States with vast oceans of international airspace to our east and west for launch and descent trajectories.

Absent a real need, or the prospect of one, it would be rash to define outer space or delimit a boundary, particularly since a likely necessary corollary, which is acceptable regime for access to outer space through an area of extended airspace claims, is more easily defined in unilateral proposals of theoretical seminars than it is diplomatically achieved and made binding on the global community of nations.

Militarization

In addition to the foregoing topics, militarization of outer space as an issue extending beyond the concern of an arms race in outer space should be noted. This issue is being raised with some persistence by a group of countries which includes Brazil, Egypt, India, Nigeria and Yugoslavia. Statements of the concern of "some delegations" about the increasing militarization of space have been included in the reports of COPUOS and its subcommittees this year. During the recent session of the Preparatory Commitee for UNISPACE '82, the global space conference to be held in Vienna in August, those delegations sought to include statements against military use of space in virtually every section of the draft conference report. These were not to be expressed as the views of some delegations, but as a consensus of the conference. To the extent the

United States and the Soviet Union have been investing in the deployment of non-weapons military satellites, we must consider them important to the protection of our security and that of our allies. It is, or should be, evident that there is no real consensus available to label military use of space as an entirely negative phenomenon, a significant obstacle to the peaceful use of outer space and to the enjoyment of the benefits of outer space by all. Nevertheless, the proponents of such language have refused to settle for a more moderate possible consensus text on the matter and have kept the subject open as one of the relatively small number of passages of the draft report on which the Preparatory Committee is not able to submit a consensus recommendation. Thus, militarization of outer space is potentially open for political confrontation at UNISPACE '82. However, it is unlikely to be transformed into an item of serious substantive negotiation on the UN's outer space law agenda.

Conclusion

While the UN's formal space law agenda does have certain implications for international security, there is not a large measure of agreement on the evaluation of those implications and on acceptable measures for dealing with them. The COPUOS Legal Subcommittee's agenda is reduced, and progress toward consensus principles or agreements is likely to remain slow.